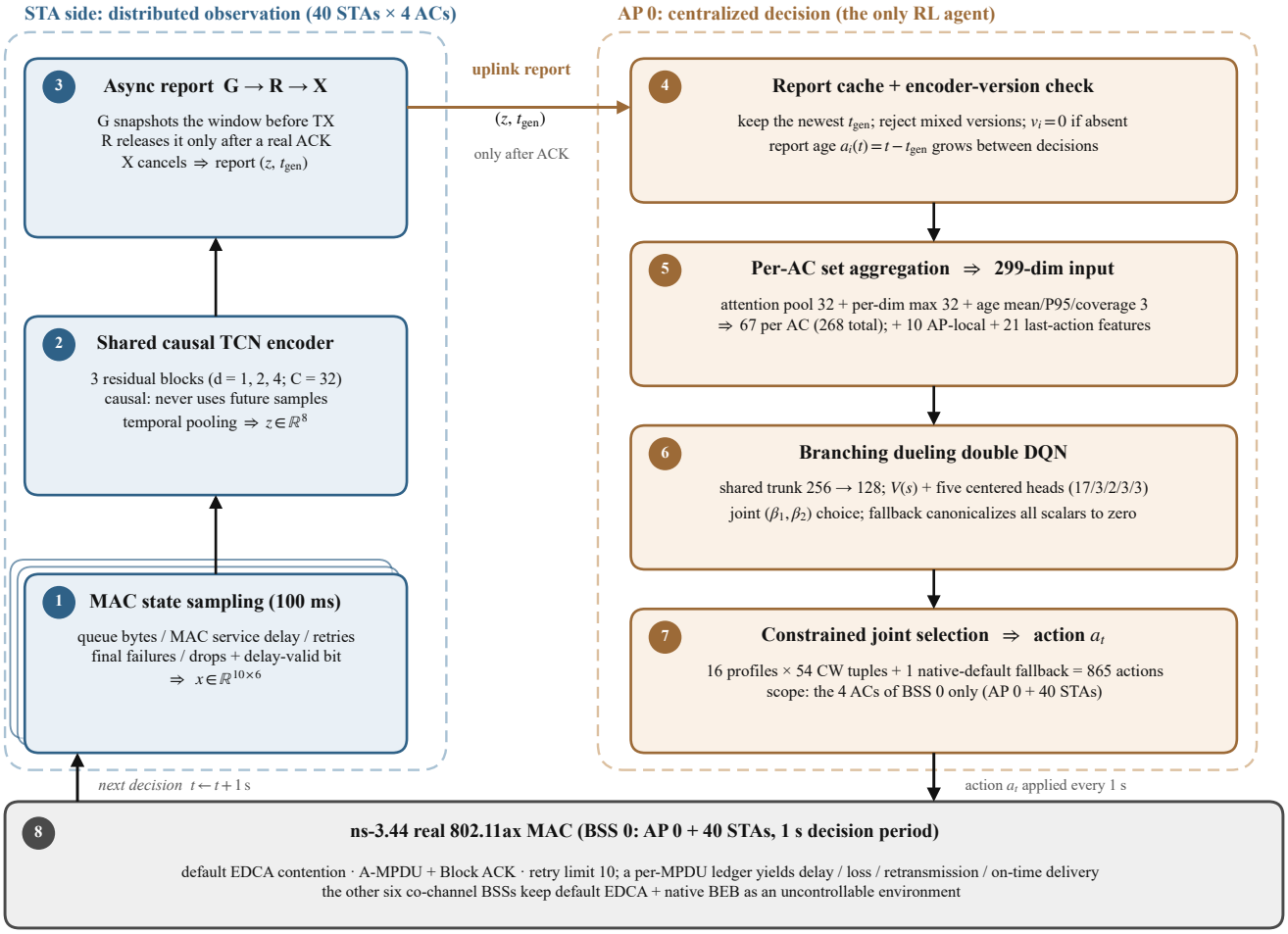

STA side: distributed observation (40 STAs × 4 ACs)
AP 0: centralized decision (the only RL agent)
3
Async report G → R → X
G snapshots the window before TX
R releases it only after a real ACK
X cancels ⇒ report (z, t_gen)
uplink report
(z, t_gen)
only after ACK
4
Report cache + encoder-version check
keep the newest t_gen; reject mixed versions; v_t = 0 if absent
report age a_i(t) = t − t_gen grows between decisions
2
Shared causal TCN encoder
3 residual blocks (d = 1, 2, 4; C = 32)
causal: never uses future samples
temporal pooling ⇒ z∈ℝ^8
5
Per-AC set aggregation ⇒ 299-dim input
attention pool 32 + per-dim max 32 + age mean/P95/coverage 3
⇒ 67 per AC (268 total); + 10 AP-local + 21 last-action features
1
MAC state sampling (100 ms)
queue bytes / MAC service delay / retries
final failures / drops + delay-valid bit
⇒ x∈ℝ^{10×6}
6
Branching dueling double DQN
shared trunk 256 → 128; V(s) + five centered heads (17/3/2/3/3)
joint (β1, β2) choice; fallback canonicalizes all scalars to zero
7
Constrained joint selection ⇒ action a_t
16 profiles × 54 CW tuples + 1 native-default fallback = 865 actions
scope: the 4 ACs of BSS 0 only (AP 0 + 40 STAs)
next decision t ← t + 1 s
action a_t applied every 1 s
8
ns-3.44 real 802.11ax MAC (BSS 0: AP 0 + 40 STAs, 1 s decision period)
default EDCA contention · A-MPDU + Block ACK · retry limit 10; a per-MPDU ledger yields delay / loss / retransmission / on-time delivery
the other six co-channel BSSs keep default EDCA + native BEB as an uncontrollable environment